Intangible Assets – Full Test Rewrite

Questions 1–21 • Answers grounded strictly in Lecture content (Lectures 7–10)
Tier-1 Analyst Protocol • No external textbook blending

Scenario A (Questions 1–7): Alex company has created a patent for a new product internally and registered it on **July 1, 2020**, at a legal cost of **$300,000**. As of that date, the patent has a **legal life of 15 years**, while Alex Co. estimated that the patent will be **useful for 10 years**. On **January 1, 2022**, the company **successfully defended** its patent by incurring legal fees of **$85,000**.

Q1 **The journal entry of recording the registration of patent on July 1, 2020:**

a. Registration expenses 300,000 (Dr) Cash 300,000 (Cr)

b. Patent 300,000 (Dr) Cash 300,000 (Cr)

c. Registration expenses 300,000 (Dr) Patent 300,000 (Cr)

d. Cash 300,000 (Dr) Patent 300,000 (Cr)

✅ **Correct Answer: b**

Why: Legal costs spent to **register** an internally created intangible asset must be **capitalized** as the asset's cost. The registration indicates the asset is economically viable and ready for use.

Evidence: *Lecture (7) p.5 – "Legal costs spent to register the internally created intangible asset should be capitalized as the registration of intangible asset is an indication that the asset is economically viable and ready for use."*

Q2 **The amortization expense of patent for year ended on December 31, 2020, =**

a. 10000

b. 15000

c. 20000

d. 30000

e. another answer

✅ **Correct Answer: b ($15,000)**

Why: Amortization = (Cost − Residual) ÷ Estimated life. Amortize over the **shorter** of useful life (10 yrs) or legal life (15 yrs) → **10 years**. 2020 = partial year from July 1 to Dec 31 = 6 months. $300,000 ÷ 10 = $30,000/yr × 6/12 = **$15,000**.

Evidence: *Lecture (7) p.5–6 – "Limited-life intangible assets are amortized over the useful life (economic life) or the legal life (contractual life) whichever is shorter." Formula: (Cost − Residual value) / Estimated life.*

Q3

The journal entry of recording the amortization expense for year ended on December 31, 2021:

a. Amortization expense 30,000 (Dr) Patent 30,000 (Cr)

b. Amortization expense 20,000 (Dr) Patent 20,000 (Cr)

c. Amortization expense 10,000 (Dr) Patent 10,000 (Cr)

d. Patent 30,000 (Dr) Amortization expense 30,000 (Cr)

✅ **Correct Answer: a**

Why: Full-year amortization for 2021 = $300,000 ÷ 10 = **$30,000**. Entry: Dr. Amortization expense, Cr. Patent (or Accumulated amortization).

Evidence: *Lecture (7) p.6 – "The amortization expense should be debited and the intangible asset or the accumulated amortization should be credited."*

Q4

The patent should be reported in the statement of financial position on December 31, 2021, by .....

a. 300,000

b. 270,000

c. 255,000

d. 240,000

✅ **Correct Answer: c ($255,000)**

Why: Carrying value = Cost − Accumulated amortization. Cost = $300,000. Accumulated amortization = 2020 ($15,000) + 2021 ($30,000) = $45,000. CV = $300,000 − $45,000 = **$255,000**.

Evidence: *Lecture (7) p.7 – "Carrying value of intangible asset = Cost − Accumulated amortization."*

Q5 **The journal entry of recording the legal fees on January 1, 2022:**

a. Cash 85,000 (Dr) Patent 85,000 (Cr)

b. Patent 85,000 (Dr) Legal fees 85,000 (Cr)

c. Legal fees 85,000 (Dr) Cash 85,000 (Cr)

d. Patent 85,000 (Dr) Cash 85,000 (Cr)

✅ **Correct Answer: d**

Why: Successful defense → legal costs are **capitalized** to the intangible asset cost. Entry: Dr. Patent (increase), Cr. Cash.

Evidence: *Lecture (7) p.7 – "If the company successfully defended the intangible asset: The legal costs should be capitalized to the intangible asset cost. Dr. Intangible asset ↑, Cr. Cash."*

Q6 The amortization expense of patent for year ended on December 31, 2022, =

a. 30000

b. 40000

c. 41750

d. another answer

✅ **Correct Answer: b ($40,000)**

Why: After successful defense, the capitalized legal cost is added to carrying value. On Jan 1, 2022: CV before defense = $255,000. Add $85,000 → new CV = **$340,000**. Remaining useful life = original 10 yrs − 1.5 yrs elapsed (Jul 2020–Dec 2021) = **8.5 years**. Amortization 2022 = $340,000 ÷ 8.5 = **$40,000**.

Evidence: *Lecture (7) p.15–16 (Example 4) – "the remaining carrying value is to be amortized over the remaining life."*

Q7

If the company failed to defend the patent on January 1, 2022, the patent should be reported in the statement of financial position on December 31, 2022, by .....

a. 300000

b. 270000

c. 255000

d. 240000

e. another answer

✅ **Correct Answer: e (another answer: $0)**

Why: If defense fails, the patent becomes **worthless**, and its remaining carrying value ($255,000) is written off as a loss. The patent balance becomes **$0** both at write-off and on Dec 31, 2022.

Evidence: *Lecture (7) p.7 – "In this case, the intangible asset becomes worthless, therefore its remaining carrying (book) value should be recorded as a loss."*

Scenario B (Questions 8–11): Green Market Inc. acquires the customer list of a large newspaper for **$360,000** on **January 1, 2020**. Green Market expects to benefit from the information over a **six-year period**, and it can sell the customer list after this period for **$60,000**. On **December 31, 2022**, Green Market management decides that this customer list is **worthless** as of that date.

Q8

In the journal entry of recording the amortization expense of customer list for year ended on December 31, 2020, the customer list should be recorded by .....

a. 50,000 (Dr)

b. 50,000 (Cr)

c. 60,000 (Dr)

d. 60,000 (Cr)

✅ **Correct Answer: b (50,000 Cr)**

Why: Amortization = ($360,000 − $60,000) ÷ 6 = **$50,000/year**. Entry: Dr. Amortization expense $50,000, **Cr. Customer list $50,000**.

Evidence: *Lecture (7) p.6 – Formula and recording: "Amortization expense … debited and the intangible asset … credited."*

Q9

In the journal entry required on December 31, 2022, the customer list should be recorded by .....

a. 180,000 (Dr)

b. 180,000 (Cr)

c. 210,000 (Dr)

d. 210,000 (Cr)

✅ **Correct Answer: d (210,000 Cr)**

Why: Carrying value at Dec 31, 2022 = Cost $360,000 − amortization for 2020, 2021, 2022 (3 × $50,000) = **$210,000**. Write-off entry: Dr. Intangible asset loss $210,000, **Cr. Customer list $210,000**.

Evidence: *Lecture (7) p.7 – "the intangible asset becomes worthless, therefore its remaining carrying (book) value should be recorded as a loss."*

Q10 The carrying value of customer list on January 1, 2023, = ...

a. 210000

b. 180000

c. 150000

d. another answer

✅ **Correct Answer: d (another answer: $0)**

Why: The asset was written off completely on Dec 31, 2022. Carrying value is **$0**. None of the listed options equals $0.

Evidence: *Lecture (7) p.7 – after write-off, the asset's remaining book value is removed.*

Q11

The amortization expense of customer list that should be reported in the income statement for year ended on December 31, 2023, = ...

a. no amortization to be recorded

b. 60000

c. 50000

d. another answer

✅ **Correct Answer: a (no amortization to be recorded)**

Why: Since the asset has been fully written off and its carrying value is $0, **no amortization** is recorded in 2023.

Evidence: *Lecture (7) p.12 – "No amortization to be recorded ... as no intangible asset has been recognized till this date." Same logic applies when the asset is derecognized.*

Scenario C (Questions 12–15): On **October 1, 2021**, Khaled company started a research and development project that aims to develop a patent for a new product and incurred costs of **$30,000** and **$60,000** in both research and development phases respectively. On **January 1, 2022**, the company decided that the R&D project became **economically viable**, so the new product was patented and registered for **10 years** at a cost of **$120,000**, while the company thinks that the patent could be useful for **12 years**.

Q12 **The journal entry of recording the research and development costs during 2021:**

a. Patent 90,000 (Dr) Cash 90,000 (Cr)

b. R&D expenses 90,000 (Dr) Cash 90,000 (Cr)

c. R&D expenses 90,000 (Dr) Patent 90,000 (Cr)

d. Patent 90,000 (Dr) R&D expenses 90,000 (Cr)

✅ **Correct Answer: b**

Why: All costs incurred **before** the point of economic viability (Jan 1, 2022) must be **expensed**. Total = $30,000 + $60,000 = $90,000. Entry: Dr. R&D expenses, Cr. Cash.

Evidence: *Lecture (7) p.4 – "Research and development costs incurred before the point of economic viability should be recorded as expenses." p.11 – "all costs incurred before the point of economic viability … should be expensed."*

Q13

The amortization expense of patent for year ended on December 31, 2021, is .....

a. 2250

b. 1875

c. 1500

d. 1250

e. another answer

✅ **Correct Answer: e (another answer: $0)**

Why: No intangible asset was recognized during 2021 (registration occurred Jan 1, 2022). Therefore, **no amortization** is recorded for 2021.

Evidence: *Lecture (7) p.12 – "No amortization to be recorded on 31/12/2023 as no intangible asset has been recognized till this date."*

Q14 **The journal entry of recording the registration cost on January 1, 2022:**

a. Patent 120,000 (Dr) Cash 120,000 (Cr)

b. Registration expenses 120,000 (Dr) Cash 120,000 (Cr)

c. Registration expenses 120,000 (Dr) Patent 120,000 (Cr)

d. Patent 120,000 (Dr) Registration expenses 120,000 (Cr)

✅ **Correct Answer: a**

Why: Registration costs incurred **after** economic viability are **capitalized** as part of the intangible asset's cost. Entry: Dr. Patent, Cr. Cash.

Evidence: *Lecture (7) p.5 – "Legal costs spent to register … should be capitalized." p.12 – Recording: Dr. Patent, Cr. Cash.*

Q15

The amortization expense of patent for year ended on December 31, 2022, = ...

a. 21000

b. 18000

c. 15000

d. 12000

e. 10000

✅ **Correct Answer: d ($12,000)**

Why: Amortize over the **shorter** of useful life (12 yrs) or legal life (10 yrs) → **10 years**. Registration date is Jan 1, 2022 → full year amortization. $120,000 ÷ 10 = **$12,000**.

Evidence: *Lecture (7) p.5–6 – "Limited-life intangible assets are amortized over the useful life … or the legal life … whichever is shorter."*

Scenario D (Questions 16–21): Presented below is information related to copyright owned by Khadija company at the **end of 2015**:
Cost: $450,000 | Carrying value: $300,000 | Fair value: $200,000 | Value in use: $250,000
As of December 31, 2015, the copyright is estimated to have a **useful life of 10 years**. At the **end of 2021**, based on new conditions in the market, the company determined the fair value and the cash flows expected from copyright to be **$120,000** and **$110,000** respectively.

Q16

The journal entry of recording the impairment of copyright on December 31, 2015 (if any):

a. Impairment loss 150,000 (Dr) Copyright 150,000 (Cr)

b. Impairment loss 100,000 (Dr) Copyright 100,000 (Cr)

c. Impairment loss 50,000 (Dr) Copyright 50,000 (Cr)

d. Copyright 100,000 (Dr) Reversal of impairment loss 100,000 (Cr)

✅ **Correct Answer: c**

Why: Recoverable amount = **higher** of Fair value ($200,000) and Value in use ($250,000) = **$250,000**. Carrying value = $300,000 > $250,000 → impairment loss = **$50,000**. Entry: Dr. Impairment loss, Cr. Copyright.

Evidence: *Lecture (8) p.2 – "Recoverable amount: the higher of Intangible asset's fair value less costs of selling … and Intangible asset's value in use." p.3–4 – "Carrying value > Recoverable amount → the asset is impaired."*

Q17 **The carrying value of copyright on January 1, 2016, = ...**

a. 300000

b. 250000

c. 200000

d. 150000

✅ **Correct Answer: b ($250,000)**

Why: Carrying value after impairment = $300,000 − $50,000 = **$250,000**.

Evidence: *Lecture (8) p.5 – "Book value at impairment test date − recognized impairment loss."*

Q18

The amortization expense of copyright for year ended on December 31, 2016, = ...

a. 40000

b. 30000

c. 25000

d. 20000

e. 15000

✅ **Correct Answer: c ($25,000)**

Why: New amortization = Book value after impairment ÷ remaining useful life. $250,000 ÷ 10 yrs = **$25,000/year**.

Evidence: *Lecture (8) p.5 – "a new amortization expense should be calculated by amortizing the book value of intangible asset after impairment over its remaining life."*

Q19

The journal entry of recording the impairment of copyright on December 31, 2021 (if any):

a. Impairment loss 10,000 (Dr) Copyright 10,000 (Cr)

b. Impairment loss 20,000 (Dr) Copyright 20,000 (Cr)

c. Copyright 10,000 (Dr) Reversal of impairment loss 10,000 (Cr)

d. Copyright 20,000 (Dr) Reversal of impairment loss 20,000 (Cr)

✅ **Correct Answer: d**

Why: By Dec 31, 2021, 6 years of amortization ($25,000 × 6 = $150,000) have been recorded. Carrying value = $250,000 − $150,000 = **$100,000**. Recoverable amount = max($120,000, $110,000) = **$120,000**. CV < Recoverable → **no new impairment**; difference of $20,000 reverses prior impairment (within original $50,000 limit). Entry: Dr. Copyright, Cr. Reversal of impairment loss.

Evidence: *Lecture (8) p.9–10 – "Carrying value < Recoverable amount … there is no impairment loss, and the difference … will be used to reverse a part of previously recognized impairment loss."*

Q20 **The carrying value of copyright on January 1, 2022, = ...**

a. 100000

b. 110000

c. 120000

d. another answer

✅ **Correct Answer: c ($120,000)**

Why: CV after reversal = $100,000 + $20,000 = **$120,000**.

Evidence: *Lecture (8) p.5 – "Book value at impairment test date + reversal of impairment loss."*

Q21

The amortization expense of copyright for year ended on December 31, 2022, = ...

a. 30000

b. 27500

c. 25000

d. 20000

✅ **Correct Answer: a ($30,000)**

Why: New amortization = CV after reversal ÷ remaining life. $120,000 ÷ 4 years (original 10 yrs − 6 yrs elapsed) = **$30,000/year**.

Evidence: *Lecture (8) p.5 – "a new amortization expense should be calculated by amortizing the book value of intangible asset after reversal of impairment loss over its remaining life."*